

CSE 460 Midterm #1

16.03.2006

Questions

1. Define the predicate **reachable**(**X**, **Y**) so that country **Y** is reachable from country **X** by car. Use the auxiliary relation **neighbor**(**X**, **Y**) that represents country **X** is the neighbor of country **Y**.
2. Let **point**(**X**, **Y**) represent a two dimensional point. Write the predicate **distance**(**X**, **Y**, **Z**) so that **Z** is the square of the distance between the two points **X** and **Y**.
3. Define the predicate **equilateral**(**X**, **Y**, **Z**) for three points **X**, **Y** and **Z**. The predicate is true if **X**, **Y** and **Z** represent three corners of an equilateral triangle.
4. Define the relation **evensum**(**L**, **Y**) which returns the sum of the even numbers in the list **L**. For example: **evensum**([3, 2, 4, 5, 12], 18).
5. Let **L1** and **L2** be two sorted lists. Write the relation **mergelist**(**L1**, **L2**, **L**) which merges those two sorted lists into another list **L**. For example: **mergelist**([1, 3, 5, 5, 7, 9], [2, 4, 6, 7, 10], [1, 2, 3, 4, 5, 5, 6, 7, 7, 9, 10]).
6. Define the relation **turkcerakamlar**(**L1**, **L2**) to translate a list of numbers between 0 and 9 to a list of the corresponding turkish words. For example: **turkcerakamlar**([3, 0, 4], [uc, sifir, dort]). Use the following as an auxiliary relation: **rakam**(0, sifir), **rakam**(1, bir), **rakam**(2, iki), ...
7. Define the predicate **strictlyincreasing**(**L**) which is true if the members of the list **L** are in strictly increasing order.
8. Define the relation **averageoflist**(**L**, **X**) so that **X** is the average of the numbers in list **L**. (Hint: Define two other relations **sumlist**(**L**, **X**), **lengthoflist**(**L**, **Y**)). For example: **averagelist**([12, 18, 24, 46], 25).
9. Define the relation **gcdlist**(**L**, **X**) where **X** is the greatest common divisor of the numbers in list **L**. For example: **gcdlist**([120, 180, 240, 48], 12).
10. Define the relation **daysinmonths**(**L1**, **L2**) so that **L1** contains the indexes of the months and **L2** is the number of days in those months. Assume February (second month has 28 days). For example: **daysinmonths**([1, 4, 2, 12], [31, 30, 28, 31]).